13	Two monoatomic gases, X and Y, are in thermal equilibrium in a mixture. The molecular mass of gas Y is half the molecular mass of gas X. The root mean square speed of the molecules of gas Y is *v*. What is the root mean square speed of the molecules of gas X?	
	A	$\frac{v}{2}$
	B	$\frac{v}{\sqrt{2}}$
	C	v
	D	$2v$
	Solution: B **Thermal equilibrium means same temperature.** **Same temperature means that the mean translational KE is the same.** **But rms speed need to be** $\frac{v}{\sqrt{2}}$ **for the mass to be doubled and KE to be still the same.** $\langle E_K \rangle = \frac{3}{2}kT$ $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$	